



College of Artificial Intelligence (El Alamein)

LYRIC LOGIC – PREDICTING SONG ARTISTS FROM LYRICS

Natural Language Processing (IN321): Final Project Proposal



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1. Project Description:

- The goal of "Lyric Logic" is to create an NLP system that accurately predicts artists from song lyrics. The ultimate aim is to deploy Lyric Logic for user applications, allowing music lovers to discover artists based on song lyrics.
- "Lyric Logic" is pivotal in Natural Language Processing (NLP), showcasing its ability to extract insights from song lyrics. It demonstrates NLP's practical applications in music, enabling personalized recommendations and enhancing human-computer interaction.
- "Lyric Logic" aims to accurately predict artists, extract their distinctive lyrical features, and measure stylistic similarities, addressing challenges in semantic understanding and stylistic analysis of song lyrics.

2. Problem Statement:

- The project aims to create an NLP system that predicts artists from song lyrics, catering to the demand for personalized music exploration.
- This problem is significant as it applies NLP techniques to meet the demand for personalized music recommendations. By predicting artists from song lyrics, it aligns with the course's focus on implementing NLP pipelines in practical applications.

3. Methodology:

- In this project, Python and Jupyter Notebook will be used as the primary programming tools. TensorFlow and Keras will serve as our main frameworks for model construction. Additionally, we'll employ NumPy, Pandas, and Matplotlib for dataset preprocessing and visualization. For text preprocessing and cleaning, we'll rely on the NLTK library. Finally, we'll use Streamlit for project deployment.
- These methods are ideal for the task due to their efficiency and widespread use in machine learning and NLP. Python and Jupyter Notebook offer simplicity and interactivity for data manipulation and analysis. TensorFlow and Keras provide powerful tools for building and training neural network models. NumPy, Pandas, and Matplotlib enable efficient data manipulation, preprocessing, and visualization. NLTK offers comprehensive tools for text preprocessing, while Streamlit simplifies project deployment by creating interactive web applications.

4. Data:

- The dataset utilized in this project is sourced from Deep Shah on Kaggle, titled "Song Lyrics Dataset".
- The dataset comprises 21 CSV files, each corresponding to a distinct artist. Within each file, there are six distinct features:
 - Artist: Artist's Name.
 - Title: Song Title.
 - Album: Album Name.
 - Date: Song Release Date.

- Lyric: Song Lyrics.
- Year: Song Release Year.

5. **Evaluation Metrics:**

- The model's performance will be evaluated during training using a subset of the dataset designated as the validation data, and subsequently after the completion of training using another subset known as the testing data. Additionally, the model's effectiveness will be assessed based on user queries within the web application.
- The evaluation of the model's performance will be based on accuracy, precision, recall, and F1-score. Accuracy will measure the overall correctness of the model's predictions. Precision will assess the proportion of correctly predicted artist instances among all instances predicted as that artist. Recall, on the other hand, will evaluate the proportion of correctly predicted artist instances among all instances of that artist in the dataset. Additionally, the F1-score will offer a balanced assessment of the model's performance, considering both the precision and recall.

6. **Expected Results:**

- To develop "Lyric Logic," a robust NLP model for accurately predicting artists from song lyrics. Offering personalized recommendations and enhance music exploration based on lyrical preferences.
- These results are vital as they directly address the project's goals. By assessing metrics like accuracy, precision, recall, and F1-score, we gain insights into the model's effectiveness in predicting artists accurately.

7. **References:**

- <https://www.kaggle.com/datasets/deepshah16/song-lyrics-dataset/data>
- <https://slolla.github.io/files/nlp.pdf>
- <https://github.com/etarakci/music-genre-prediction>